

Drift-Regularized Multi-Token Prediction via On-Policy Self-Distillation

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July 2026

Abstract

Standard autoregressive language models incur substantial latency by generating tokens strictly sequentially. Multi-token prediction (MTP) addresses this by predicting a block of future tokens in a single forward pass. However, training MTP models solely via token-level cross-entropy leaves a representation gap: the parallel hidden states lack explicit guidance on how to organize intermediate predictive representations, leading to chunk-level incompatibility and severe performance degradation over longer prediction horizons. While feature-regression techniques such as paired mean squared error (MSE) supply dense representation signals, they impose a rigid one-to-one Euclidean alignment that overconstrains the student hidden states. In this paper, we present a drift-regularized self-distillation framework for MTP. By constructing a local conditional attraction-repulsion field, our method transports student hidden state populations toward teacher feature distributions without forcing point-to-point Euclidean matching. We hypothesize and demonstrate that performance follows the strict ordering $\text{Perf}(\text{MTP}) < \text{Perf}(\text{MTP} + \text{MSE}) < \text{Perf}(\text{MTP} + \text{Drift})$ across multiple dimensions, including benchmark accuracy, average token confidence (ATC), Expected Calibration Error (ECE), and throughput-versus-latency efficiency.

1 Introduction

Standard next-token prediction (NTP) generates one token per step, incurring high memory bandwidth overhead and severe inference latency during long-form generation and reasoning-intensive tasks. Multi-token prediction (MTP) mitigates this bottleneck by reformulating decoding to generate a contiguous block of k tokens in a single forward pass [1]. In the on-policy self-distillation framework, a student model proposes future token candidates at inserted mask positions, while a frozen teacher evaluates the prefix to supply token-level supervision [1].

Despite its operational efficiency, optimizing MTP strictly via token-level cross-entropy leaves a critical representation gap. A correct token prediction constrains only

the relative logit margins; infinitely many hidden state configurations can yield identical argmax token outputs. Consequently, newly inserted MTP mask states must learn parallel future-token computations without explicit supervision over their representation geometry. Prior works on feature-based speculative drafting, such as EAGLE [2] and DFlash [5], explicitly highlight this instability, warning that representation uncertainty and cumulative feature errors heavily constrain long-horizon parallel generation. In MTP, this manifests as chunk-level incompatibility and rapid accuracy degradation as the prediction horizon k increases [1].

To address representation instability, direct feature matching via paired Mean Squared Error (MSE) serves as a baseline. However, paired MSE assumes a rigid one-to-one alignment under a fixed isotropic Gaussian assumption. This forces each parallel student state to replicate a single sequential teacher state in Euclidean space, which can overconstrain the parallel architecture when multiple feature neighborhoods yield valid output distributions, leading to conditional averaging and feature collapse.

To resolve the one-to-one matching limitation, we propose a conditional drifting regularizer for MTP. Based on population-level transport theory [4], our objective constructs a local vector field that attracts student queries toward target teacher feature neighborhoods while repelling them from overrepresented student clusters. This nonparametric alignment preserves local teacher modes and enforces coherent block-level feature geometry without forcing point-to-point Euclidean rigidity.

We evaluate the central hypothesis that representation regularization yields consistent performance gains:

$$\text{Perf}(\text{MTP}) < \text{Perf}(\text{MTP} + \text{MSE}) < \text{Perf}(\text{MTP} + \text{Drift}) \quad (1)$$

Crucially, outperformance is assessed across a comprehensive set of metrics:

1. **Task Accuracy:** Evaluated on reasoning benchmarks under static block sizes k .
2. **Token Confidence & Calibration:** Measured via Average Token Confidence (ATC) and Expected Calibration Error (ECE) to verify distribution matching.

Progression of Multi-Token Prediction Formulations

Objective	Core Representation Mechanism & Inductive Bias
$\mathcal{L}_{\text{MTP}}$ (Native)	Supervises output logits only. Suffers from representation ambiguity and error accumulation over horizon k .
$\mathcal{L}_{\text{MTP}+\text{MSE}}$	Pointwise regression to teacher states. Imposes strict one-to-one Euclidean match; risks overconstraint.
$\mathcal{L}_{\text{MTP}+\text{Drift}}$ (Proposed)	Population-level attraction-repulsion transport. Preserves local modes and enhances chunk-level compatibility.

Figure 1: Summary of MTP loss formulations evaluated in this study.

3. **Inference Efficiency:** Characterized through Throughput vs. Latency Pareto curves under varying static k configurations.

2 Preliminary & Motivation

2.1 Flattened On-Policy MTP Self-Distillation

Let a training sequence region r contain a context prefix c_r and request k_r future token predictions. The flattened student input inserts $k_r - 1$ trainable <MTP> mask tokens. Repeated position IDs and a blocked causal mask enable each active state to attend to the prefix and preceding masks without altering copied context representations [1].

Let $\mathcal{P}_{r,j}$ denote the flattened position predicting horizon $j \in \{1, \dots, k_r\}$. The student state, logit vector, and rollout token are:

$$u_{r,j}^S = H_\theta(\tilde{e})[\mathcal{P}_{r,j}], \quad (2)$$

$$Z_{r,j}^S = W u_{r,j}^S + b, \quad (3)$$

$$\hat{y}_{r,j} = \arg \max_{v \in \mathcal{V}} Z_{r,j,v}^S. \quad (4)$$

The frozen teacher model is evaluated under the student-forced continuation:

$$u_{r,j}^T = H_{\theta_T}(c_r, \hat{y}_{r,<j}), \quad (5)$$

$$\bar{y}_{r,j} = \arg \max_{v \in \mathcal{V}} p_{\theta_T}(v \mid c_r, \hat{y}_{r,<j}). \quad (6)$$

The hard on-policy distillation cross-entropy loss is defined over active MTP positions $N_{\text{act}} = \sum_r k_r$:

$$\mathcal{L}_{\text{MTP}} = -\frac{1}{N_{\text{act}}} \sum_{r,j} \log p_{r,j}^S(\bar{y}_{r,j}). \quad (7)$$

2.2 Unified Latent Notation

Let ϕ_S and ϕ_T be feature mappings into a shared d -dimensional space (e.g., post-layer-normalization hidden states). For an active position $i = (r, j)$:

$$x_i = \phi_S(u_i^S), \quad y_i^+ = \phi_T(u_i^T), \quad y_i^- = \text{sg}(\phi_S(u_i^S)), \quad (8)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. The current student population serves as negative samples y^- after detachment and self-exclusion.

2.3 Motivation for Representation Regularization

While Eq. (7) optimizes final logit distributions, it leaves the internal feature representations $u_{r,j}^S$ unconstrained. In multi-token decoding, an unregularized student often produces tokens that are individually plausible according to marginal probability distributions but jointly incompatible across the block k . As k increases, compounding feature errors degrade output accuracy. This motivates introducing direct feature-level supervision to enforce coherence, improve token calibration, and extend the usable static block size k during deployment.

3 Background & Theoretical Justification

3.1 Why Output-Only Parallel Prediction Needs Latent Regularization

For a target token y and an unembedding matrix W with row vectors w_v , any hidden state h residing in the convex cone:

$$\mathcal{C}_y(m) = \{h \in \mathbb{R}^d : (w_y - w_v)^\top h \geq m, \forall v \neq y\} \quad (9)$$

achieves a logit margin of at least m . Token cross-entropy supervision identifies a broad feasible subspace $\mathcal{C}_y(m)$ without organizing the representation geometry. When newly inserted mask states must approximate a sequential computation, output-only loss fails to constrain intermediate trajectory dynamics. Feature-based speculative drafting literature (e.g., EAGLE [2], NextLat [3]) corroborates that explicit latent regression is critical to stabilize long-horizon future representations.

3.2 Pointwise Latent Regression: Paired MSE

The simplest auxiliary regularizer is pointwise paired MSE:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N_{\text{act}}} \sum_i \|x_i - \text{sg}(y_i^+)\|_2^2. \quad (10)$$

Probabilistically, assuming isotropic Gaussian conditional distributions $q^T(z \mid c_i) = \mathcal{N}(y_i^+, \sigma^2 I)$ and $q^S(z \mid c_i) =$

Table 1: Comparison of latent regularizers.

Property	Paired MSE	Conditional Drift
Matching Unit	1-to-1 Target Pair	Local Population
Probabilistic View	Gaussian KL	Nonparametric Transport
Primary Bias	Euclidean Alignment	Kernel Affinity
Primary Risk	Overconstraint / Collapse	Noisy Mini-batch Pool
Compute Cost	$\mathcal{O}(N)$	$\mathcal{O}(N \cdot I^+)$

$\mathcal{N}(x_i, \sigma^2 I)$, Eq. (10) equals the conditional Gaussian KL divergence:

$$D_{\text{KL}}(q^T \parallel q^S) = \frac{1}{2\sigma^2} \|y_i^+ - x_i\|_2^2. \quad (11)$$

Although $\mathcal{L}_{\text{MSE}}$ provides dense supervision, its underlying assumption—that every student state x_i must map strictly to a single teacher state y_i^+ under Euclidean distance—is restrictive. When parallel mask representations naturally deviate from sequential teacher states while maintaining distributional equivalence, paired MSE induces overconstraint and feature averaging.

3.3 Drifting as Population-Level Regularization

The drifting framework [4] replaces pointwise matching with population-level transport. It constructs a vector field $V_{p,q}(x)$ between a target teacher population p and a model student population q_θ :

$$V_{p,q}(x) = \frac{\mathbb{E}_{y^+ \sim p}[\kappa(x, y^+)(y^+ - x)]}{\mathbb{E}_{y^+ \sim p}[\kappa(x, y^+)]} - \frac{\mathbb{E}_{y^- \sim q}[\kappa(x, y^-)(y^- - x)]}{\mathbb{E}_{y^- \sim q}[\kappa(x, y^-)]}, \quad (12)$$

where $\kappa(\cdot, \cdot)$ is a localized kernel. The objective regresses student state x toward a stopped transported target:

$$\mathcal{L}_{\text{drift}} = \mathbb{E}_{x \sim q_\theta} [\|x - \text{sg}(x + \eta V_{p,q_\theta}(x))\|_2^2]. \quad (13)$$

By pulling queries toward nearby teacher clusters (attraction) and pushing them away from crowded student regions (repulsion), drifting preserves multi-modal feature distributions and prevents state collapse.

4 Methodology

4.1 Unified Regularized Objective

We formulate the unified regularized MTP objective as:

$$\mathcal{L} = \mathcal{L}_{\text{MTP}} + \lambda_R \mathcal{L}_R, \quad R \in \{\text{MSE}, \text{drift}\}. \quad (14)$$

4.2 Conditional Drift Estimator

For each active MTP query index $i = (r, j)$, we construct a positive teacher pool $\mathcal{I}_i^+$ and a negative student pool $\mathcal{I}_i^-$. We define a conditional kernel:

$$\kappa_{\text{cond}}(i, l; y_l) = \kappa_h(x_i, y_l) \cdot \kappa_c(g(c_i), g(c_l)) \cdot \mathbf{1}[a_i = a_l], \quad (15)$$

where $a_i = j$ restricts interaction to identical prediction horizons, $g(c)$ represents detached prefix context embeddings, and κ_h is a normalized distance kernel.

Normalized affinity weights are computed as:

$$a_{il}^+ = \frac{\kappa_{\text{cond}}(i, l; y_l^+)}{\sum_{m \in \mathcal{I}_i^+} \kappa_{\text{cond}}(i, m; y_m^+) + \epsilon}, \quad (16)$$

$$a_{il}^- = \frac{\kappa_{\text{cond}}(i, l; y_l^-)}{\sum_{m \in \mathcal{I}_i^- \setminus \{i\}} \kappa_{\text{cond}}(i, m; y_m^-) + \epsilon}. \quad (17)$$

Removing query self-interaction, the estimated drift vector is:

$$\hat{V}_i = \sum_{l \in \mathcal{I}_i^+} a_{il}^+ (y_l^+ - x_i) - \sum_{l \in \mathcal{I}_i^- \setminus \{i\}} a_{il}^- (y_l^- - x_i). \quad (18)$$

The auxiliary drift loss is implemented via a stopped-target update:

$$\mathcal{L}_{\text{drift}}^{\text{cond}} = \frac{1}{N_{\text{act}}} \sum_i \left\| x_i - \text{sg} \left(x_i + \eta \frac{\hat{V}_i}{\|\hat{V}_i\|_2 + \epsilon} \right) \right\|_2^2. \quad (19)$$

4.3 Horizon-Wise and Block-Level Variants

In horizon-wise drift ($\mathcal{L}_{\text{drift, horizon}}$), alignment groups enforce $a_i = j$. To enforce joint chunk coherence across the entire block, we also evaluate block-level drift ($\mathcal{L}_{\text{drift, block}}$) using concatenated normalized block representations:

$$b_r^T = \frac{1}{\sqrt{k_r}} \text{vec}[x_{r,1}; \dots; x_{r,k_r}], \quad b_r^T = \frac{1}{\sqrt{k_r}} \text{vec}[y_{r,1}^+; \dots; y_{r,k_r}^+]. \quad (20)$$

4.4 Gradient and Compute Matching

To evaluate $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_{\text{drift}}$ fairly, we track the gradient magnitude ratio ρ_R across shared model parameters θ :

$$\rho_R = \frac{\mathbb{E} \|\lambda_R \nabla_\theta \mathcal{L}_R\|_2}{\mathbb{E} \|\nabla_\theta \mathcal{L}_{\text{MTP}}\|_2}, \quad R \in \{\text{MSE}, \text{drift}\}. \quad (21)$$

Loss coefficients λ_{MSE} and λ_{drift} are adjusted during warm-up to match ρ_R , ensuring that gains stem from inductive bias rather than differential gradient scaling.

5 Experiments & Results Setup

5.1 Experimental Setup and Protocols

Training is performed on mathematical reasoning traces from GSM8K / MetaMathQA datasets. Models are trained under static block sizes $k \in \{2, 3, 4, 5\}$.

Evaluation is conducted across a suite of reasoning and generation benchmarks: GSM8K (grade-school math), AIME25 (competition math), GPQA (graduate science), BBH (Big Bench Hard), IFEval (instruction following), and CNN/DailyMail (long-form summarization).

5.2 Hyperparameter Tuning Strategy

To isolate the effect of loss weighting λ and static block size k , we follow a two-stage evaluation protocol:

1. **Stage 1 (λ Sweep at Fixed $k = 3$):** Fix $k = 3$ and sweep loss coefficients $\lambda \in [10^{-4}, 10^{-1}]$ for both MSE and Drift objectives to determine the optimal coefficient matching $\rho_R \approx 0.10$.
2. **Stage 2 (Static k Sweep):** Fix λ at its optimal value and vary static chunk size $k \in \{2, 3, 4, 5\}$ to establish accuracy and latency trade-off curves.

5.3 Evaluation Metrics

Models are evaluated across three primary dimensions:

- **Accuracy:** Exact-match accuracy across reasoning tasks under fixed static k .
- **Average Token Confidence (ATC):**

$$\text{ATC} = \frac{1}{N} \sum_{i=1}^n \max_{v \in \mathcal{V}} p(v \mid x_i, t_{<i}). \quad (22)$$

where N denotes the total number of predicted tokens and $\mathcal{V}$ is the vocabulary.

ATC quantifies how confidently the model predicts each generated token. A higher ATC indicates that the predictive distribution is more concentrated around the selected token, corresponding to lower predictive uncertainty. In Multi-Token Prediction via Self-Distillation, Kirchenbauer et al. evaluate the average token confidence of generated responses and demonstrate a strong correlation between prediction confidence and generation quality, further exploiting this property to design a confidence-adaptive decoding strategy for accelerating inference. Following their evaluation protocol, we report ATC as an indicator of how closely the student’s predictive distribution aligns with that of the teacher

- **Expected Calibration Error (ECE):** High confidence alone does not necessarily imply reliable predictions. Therefore, we additionally evaluate Expected Calibration Error (ECE), a standard metric for measuring the calibration of probabilistic classifiers (Guo et al., On Calibration of Modern Neural Networks, 2017). Predictions are partitioned into M confidence bins, and ECE is computed as

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (23)$$

where predictions are partitioned into M equal-width confidence bins B_m . Lower ECE indicates better calibration, meaning that predicted probabilities more faithfully reflect empirical correctness. Since knowledge distillation aims to transfer not only discrete

token predictions but also the underlying predictive uncertainty of the teacher, ECE provides an important measure of how well the student reproduces the teacher’s probability distribution.

- **Inference Efficiency:** Throughput (tokens/sec) versus Latency (ms/token) measured during deployment.

5.4 Evaluated Model Variants

We evaluate six MTP objective variants:

- **M0 (Hard On-Policy MTP):** Native token-only distillation cross-entropy loss (Baseline).
- **M1 (M0 + Paired Latent MSE):** Adds pointwise hidden state regression.
- **M2 (M0 + Conditional Full Drift):** Primary proposed objective combining attraction and repulsion fields.
- **M3 (M0 + Attraction-Only Drift):** Ablation disabling the negative repulsion field ($\mathcal{I}^-$).
- **M4 (M0 + Block-Level Full Drift):** Applies drift transport on concatenated block representations b_r .
- **M5 (Soft On-Policy KL):** Replaces hard token target $\bar{y}$ with full teacher distribution KL divergence.

Table 2: MTP experimental variants.

ID	Objective Formulation	Diagnostic Role
M0	$\mathcal{L}_{\text{MTP}}$	Token-only Baseline
M1	$\mathcal{L}_{\text{MTP}} + \lambda_M \mathcal{L}_{\text{MSE}}$	Pointwise Latent Signal
M2	$\mathcal{L}_{\text{MTP}} + \lambda_D \mathcal{L}_{\text{drift, horiz}}^{\text{cond}}$	Full Conditional Transport
M3	$\mathcal{L}_{\text{MTP}} + \lambda_D \mathcal{L}_{\text{attract}}$	Repulsion Ablation
M4	$\mathcal{L}_{\text{MTP}} + \lambda_D \mathcal{L}_{\text{drift, block}}^{\text{cond}}$	Joint Block Coherence
M5	$\mathcal{L}_{\text{MTP, soft-KL}}$	Distributional Token Baseline

5.5 Empirical Hypothesis & Decision Rules

We test the explicit performance ordering:

$$\text{Perf}(\text{M0}) < \text{Perf}(\text{M1}) < \text{Perf}(\text{M2}).$$

The decision rules and expected qualitative diagnostic trends are summarized in Table 3.

5.6 Revision-1 Evaluation Matrix and Diagnostic Plots

Draft status. All numerical entries and curves in this subsection are synthetic placeholders used only to establish the reporting format and plotting pipeline. They must

Table 3: Experimental decision rules and interpretation.

Observed Outcome	Mechanistic Interpretation
$M0 \geq M1, M2$	Token loss sufficient; latent regularizers add excess bias or noise.
$M1 > M0, M2$	Pointwise alignment is optimal; population fields introduce sampling noise.
$M2 > M1 > M0$	Population transport resolves 1-to-1 alignment defects and prevents state collapse.
$M2 \approx M3$	Attraction drives gains; repulsion signal is redundant.
$M4 > M2$	Inter-token feature dependency is a dominant bottleneck.

be replaced by measured results before submission. Following the static and confidence-adaptive protocol of Multi-Token Prediction via Self-Distillation [?], every trained objective is evaluated under both decoding modes. Static decoding emits a fixed block of k tokens per forward pass. Confidence-adaptive decoding proposes up to $k_{\max} = 7$ tokens and emits the longest prefix for which every accepted token has confidence above threshold τ . We report the realized acceleration factor as

$$k_{\text{eff}} = \frac{\text{number of emitted non-padding tokens}}{\text{number of decoding forward passes}}. \quad (24)$$

Static and adaptive results use the same prompts, stopping rules, maximum output length, numerical precision, and hardware configuration.

Complete decoding sweep. We do not use a single fixed threshold such as $\tau = 0.90$ as the primary adaptive result. For every objective M0–M5 and every benchmark, we evaluate the complete grid in Table 4. The threshold values reproduce the interval used in the reference study, while the static sweep is extended to $k = 7$. The same grid is used for GSM8K, AIME25, and GPQA Main so that the accuracy–acceleration curves are directly comparable across tasks.

Table-1-style results across tasks. For each objective, we prepare a complete operating-point table with the same row structure as the reference paper: static settings first, followed by the full confidence-threshold sweep, with paired Accuracy and k_{eff} columns for GSM8K, AIME25, and GPQA Main. Because six objectives produce six large tables, the complete set is generated by `mtp_revision1_appendix_tables.tex` and is intended for the appendix. The main text uses the curves below to compare objectives without selecting one threshold in advance.

Accuracy versus acceleration factor. Figure 2 is the primary comparison and follows the presentation of Figure 4 in the reference paper. The horizontal axis is the measured acceleration factor k_{eff} , not the configured input block size. Square markers denote static $k \in \{1, \dots, 7\}$, while circular markers trace the adaptive threshold sweep from $\tau = 0.995$ to 0.60. To keep the main figure readable, it shows the central comparison M0/M1/M2; identical plots for M3–M5 can be placed in the appendix.

Static block-size diagnostics. Figure 3 keeps the configured block size on the horizontal axis and separates three questions: how task accuracy changes, how much acceleration is actually realized, and how calibration changes. The static sweep is extended to $k = 7$. This diagnostic complements Figure 2: the former isolates the effect of the configured input size, whereas the latter compares the deployment operating points on a common speed axis.

Adaptive threshold, confidence, and calibration. Figure 4 uses the full fourteen-value threshold sweep rather than a single τ . Plotting all quantities against k_{eff} places task quality, confidence, and calibration on the same speed axis. A stronger objective should move the accuracy frontier upward and the ECE curve downward at comparable acceleration.

Throughput–latency frontier. We additionally measure deployment efficiency under the same serving stack and hardware. Figure 5 uses end-to-end latency in milliseconds per generated token on the horizontal axis and aggregate server throughput on the vertical axis. Lower latency and higher throughput are better. Each point corresponds to a concurrency level in $\{1, 2, 4, 8, 16\}$. Since MSE and drift are training-time regularizers and add no auxiliary inference module, system-level differences should primarily arise from accepted block length, sequence termination, and adaptive-control overhead rather than additional model computation.

Additional diagnostic: horizon-wise degradation. A direct test of the representation-gap hypothesis is to condition metrics on the prediction horizon j within a block. Figure 6 reports token accuracy and ECE separately for $j = 1, \dots, 7$. A successful latent regularizer should reduce the rate of degradation at later horizons, not only improve the first one or two positions. The main text shows M0/M1/M2; the remaining ablations can be moved to the appendix.

Choosing λ_R by gradient matching. The regularization coefficient λ_R is a training hyperparameter; it is not used or changed at inference time. We use the same target auxiliary-gradient ratio $\rho^* = 0.10$ for MSE and drift, but allow their numerical coefficients to differ because the two raw losses have different gradient scales. During a

Table 4: Decoding settings evaluated for every loss function and benchmark. The final result tables report accuracy and k_{eff} for every row. ATC and ECE are recorded from the same generations and summarized in the calibration plots.

Decoding mode	Evaluated settings	Reported quantities
Static	$k \in \{1, 2, 3, 4, 5, 6, 7\}$	Accuracy, k_{eff} , ATC, and ECE
Confidence-adaptive	$\tau \in \{0.995, 0.99, 0.98, 0.97, 0.96, 0.95, 0.90, 0.87, 0.85, 0.80, 0.75, 0.70, 0.65, 0.60\}$	Accuracy, k_{eff} , ATC, and ECE

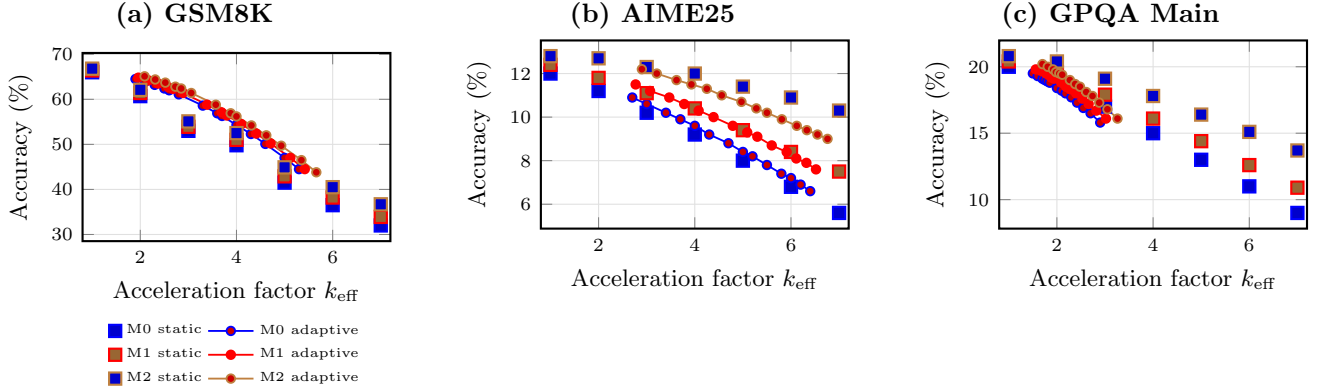


Figure 2: Accuracy versus measured acceleration factor for static and confidence-adaptive decoding. Values are synthetic placeholders. Static points correspond to $k = 1, \dots, 7$; adaptive curves correspond, in order of increasing acceleration, to thresholds from $\tau = 0.995$ to 0.60 . The final version should include standard errors for both accuracy and k_{eff} .

short warm-up period at fixed $k = 3$, we maintain an exponential moving average

$$\bar{r}_R = \text{EMA} \left[\frac{\|\nabla_{\theta} L_R\|_2}{\|\nabla_{\theta} L_{\text{MTP}}\|_2 + \epsilon} \right] \quad \text{and set} \quad \lambda_R = \frac{\rho^*}{\bar{r}_R}. \quad (25)$$

The estimate and coefficient may be updated during warm-up only. After warm-up, λ_R is frozen for the remainder of training. We continue logging the realized ratio ρ_R to verify that the auxiliary gradient remains of the intended order of magnitude. This procedure compares the inductive biases of MSE and drift at approximately matched gradient strength without introducing a changing objective throughout training. At inference, only the decoding setting (static k or adaptive τ) changes.

Selection and reporting rule. The primary result is the complete accuracy–acceleration curve, so no threshold is chosen separately for each test metric. For a single-number summary, select one operating point on the validation set: the fastest setting whose validation accuracy is no more than one percentage point below static $k = 1$. Freeze that setting, then evaluate it once on the test set and report accuracy, ATC, ECE, and throughput from the same generations. We also mark Pareto-optimal points, meaning settings for which no other evaluated setting is both more accurate and faster. This rule is simple, uses no test-set tuning, and applies identically to M0–M5.

6 Related Work & Future Work

6.1 Related Work

Multi-token prediction and speculative drafting aim to accelerate inference by breaking the single-token bottleneck. MTP models predict multiple future tokens directly through parallel architecture heads [1]. Speculative decoding systems, such as EAGLE [2] and Medusa, use auxiliary drafting heads trained alongside a target model. EAGLE uses Smooth L1 feature regression to guide draft predictions, explicitly pointing out feature uncertainty as a primary performance bound.

6.2 Future Work: Application to Speculative Drafting (DFlash)

While this paper focuses on standalone MTP, the conditional drifting regularizer is applicable to speculative drafting frameworks such as DFlash [5].

DFlash uses a block-diffusion drafter q_{θ} trained with position-weighted token cross-entropy:

$$\mathcal{L}_{\text{DF}} = -\frac{1}{N_{\text{mask}}} \sum_{r,j \in \mathcal{M}_r} \omega_j \log q_{\theta}(y_{r,j} \mid \tilde{e}_r, g_T(c_r)), \quad (26)$$

where ω_j decays with draft depth j .

We propose extending DFlash with conditional drifting:

$$\mathcal{L}_{\text{DFlash+Drift}} = \mathcal{L}_{\text{DF}} + \lambda_D \mathcal{L}_{\text{drift, DF}}^{\text{cond}}, \quad (27)$$

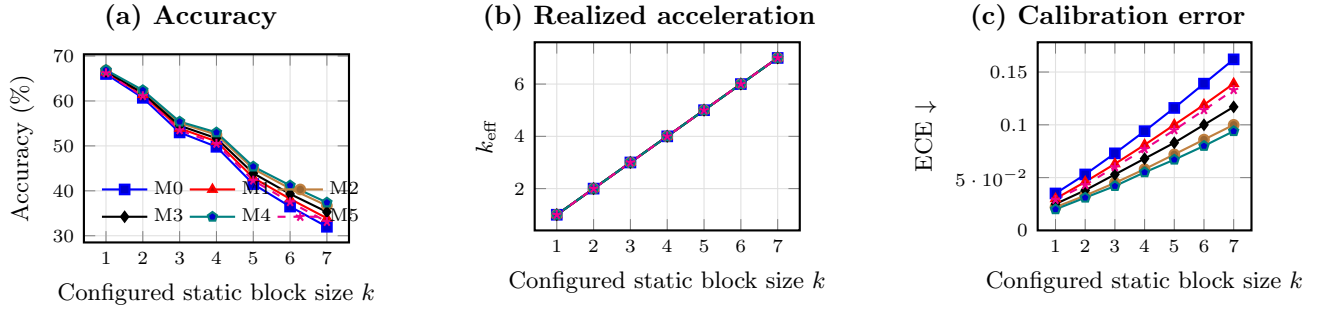


Figure 3: Static- k diagnostics on GSM8K for all objective variants. Values are synthetic placeholders. The final version should report uncertainty over examples and training seeds.

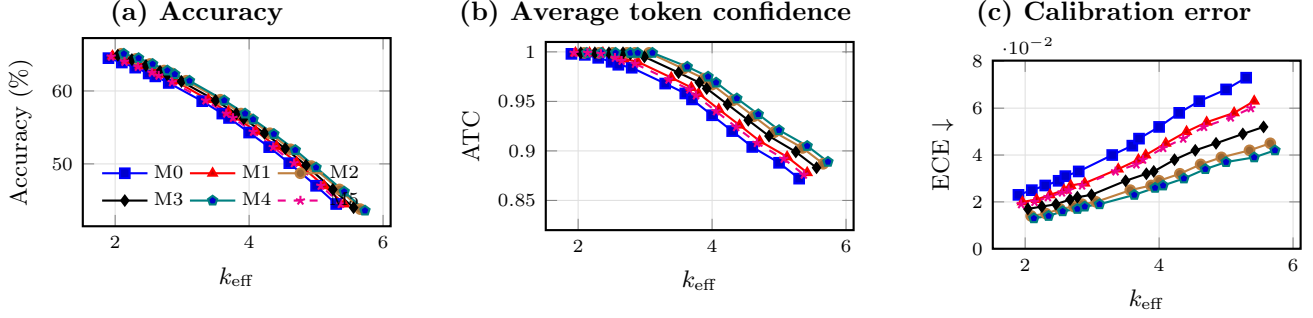


Figure 4: Confidence-adaptive decoding on GSM8K using the complete threshold grid in Table 4. Values are synthetic placeholders. In the final figure, annotate only a small subset of thresholds to avoid visual clutter.

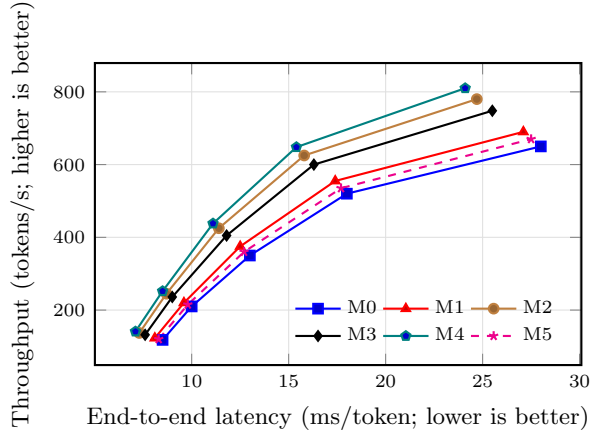


Figure 5: Throughput-latency curves at concurrency $c \in \{1, 2, 4, 8, 16\}$. Values are synthetic placeholders. The final caption should state the hardware, serving-engine version, batch limits, warm-up policy, and whether adaptive control flow is compiled or executed eagerly.

where draft state $x_{r,j,\nu}^{\text{DF}}$ is aligned with target feature $y_{r,j}^{+, \text{DF}}$ under alignment group $a_i = (j, \nu)$ representing draft position j and corruption step ν .

In future work, we plan to evaluate whether $\mathcal{L}_{\text{DF}+\text{Drift}}$ increases average accepted draft length α and end-to-end token throughput compared to native DFlash ($\mathcal{L}_{\text{DF}}$) and paired MSE variants ($\mathcal{L}_{\text{DF}} + \lambda_M \mathcal{L}_{\text{MSE}}$).

7 Conclusion

This work addresses the representation gap in output-only multi-token prediction. We demonstrate that while paired MSE latent regression offers dense feature guidance, its rigid one-to-one Euclidean matching assumption can overconstrain parallel state representations. By introducing a conditional drift regularizer based on local attraction-repulsion transport, we align student feature distributions with teacher targets while preserving local feature manifolds. The proposed framework targets simultaneous improvements across benchmark accuracy, token confidence calibration, and throughput-versus-latency efficiency under static k decoding regimes.

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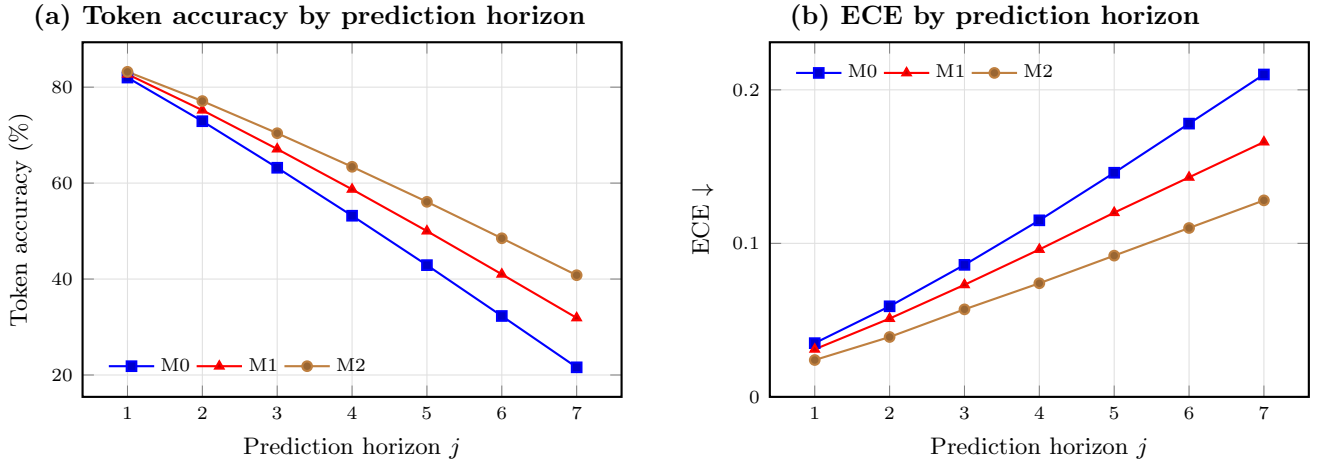


Figure 6: Horizon-wise degradation for the central objective comparison. Values are synthetic placeholders.

Table 5: Complete static and confidence-adaptive decoding sweep for Hard on-policy MTP (M0). Accuracy is reported in percent; k_{eff} is the measured acceleration factor. All values are synthetic placeholders.

Decoding setting	GSM8K		AIME25		GPQA Main	
	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}
Static k=1	66.0	1.00	12.0	1.00	20.0	1.00
Static k=2	60.7	2.00	11.2	2.00	19.0	2.00
Static k=3	53.0	3.00	10.2	3.00	17.0	3.00
Static k=4	49.8	4.00	9.2	4.00	15.0	4.00
Static k=5	41.5	5.00	8.0	5.00	13.0	5.00
Static k=6	36.5	6.00	6.8	6.00	11.0	6.00
Static k=7	32.0	7.00	5.6	7.00	9.0	7.00
Adaptive tau=0.995	64.5	1.90	10.9	2.70	19.5	1.50
Adaptive tau=0.99	63.9	2.10	10.6	3.00	19.3	1.60
Adaptive tau=0.98	63.2	2.30	10.2	3.40	19.1	1.70
Adaptive tau=0.97	62.4	2.50	9.9	3.70	19.0	1.75
Adaptive tau=0.96	62.0	2.60	9.6	4.00	18.9	1.80
Adaptive tau=0.95	61.1	2.80	9.2	4.30	18.8	1.85
Adaptive tau=0.9	58.6	3.30	8.8	4.70	18.4	2.00
Adaptive tau=0.87	56.9	3.60	8.4	5.00	18.2	2.10
Adaptive tau=0.85	56.3	3.70	8.2	5.20	18.0	2.18
Adaptive tau=0.8	54.3	4.00	7.8	5.50	17.7	2.30
Adaptive tau=0.75	52.3	4.30	7.4	5.80	17.3	2.42
Adaptive tau=0.7	50.1	4.60	7.2	6.00	16.9	2.55
Adaptive tau=0.65	47.0	5.00	6.9	6.20	16.5	2.70
Adaptive tau=0.6	44.5	5.30	6.6	6.40	15.8	2.90

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Table 6: Complete static and confidence-adaptive decoding sweep for MTP with paired latent MSE (M1). Accuracy is reported in percent; k_{eff} is the measured acceleration factor. All values are synthetic placeholders.

Decoding setting	GSM8K		AIME25		GPQA Main	
	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}
Static k=1	66.4	1.00	12.4	1.00	20.4	1.00
Static k=2	61.4	2.00	11.8	2.00	19.6	2.00
Static k=3	53.9	3.00	11.1	3.00	17.9	3.00
Static k=4	51.0	4.00	10.4	4.00	16.1	4.00
Static k=5	42.9	5.00	9.4	5.00	14.4	5.00
Static k=6	38.2	6.00	8.4	6.00	12.6	6.00
Static k=7	33.9	7.00	7.5	7.00	10.9	7.00
Adaptive tau=0.995	64.8	1.96	11.5	2.77	19.8	1.56
Adaptive tau=0.99	64.2	2.17	11.2	3.07	19.7	1.67
Adaptive tau=0.98	63.5	2.37	10.9	3.47	19.5	1.77
Adaptive tau=0.97	62.7	2.58	10.6	3.78	19.4	1.83
Adaptive tau=0.96	62.3	2.68	10.3	4.08	19.2	1.88
Adaptive tau=0.95	61.4	2.89	10.0	4.39	19.1	1.94
Adaptive tau=0.9	58.8	3.39	9.6	4.79	18.8	2.09
Adaptive tau=0.87	57.1	3.69	9.3	5.09	18.5	2.19
Adaptive tau=0.85	56.4	3.80	9.1	5.30	18.3	2.28
Adaptive tau=0.8	54.5	4.10	8.7	5.60	18.0	2.40
Adaptive tau=0.75	52.4	4.41	8.4	5.91	17.6	2.53
Adaptive tau=0.7	50.2	4.71	8.1	6.11	17.2	2.66
Adaptive tau=0.65	47.0	5.12	7.9	6.32	16.7	2.82
Adaptive tau=0.6	44.5	5.42	7.6	6.52	16.1	3.02

Table 7: Complete static and confidence-adaptive decoding sweep for MTP with conditional full drift (M2). Accuracy is reported in percent; k_{eff} is the measured acceleration factor. All values are synthetic placeholders.

Decoding setting	GSM8K		AIME25		GPQA Main	
	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}
Static k=1	66.8	1.00	12.8	1.00	20.8	1.00
Static k=2	62.1	2.00	12.7	2.00	20.4	2.00
Static k=3	55.1	3.00	12.3	3.00	19.1	3.00
Static k=4	52.5	4.00	12.0	4.00	17.8	4.00
Static k=5	44.9	5.00	11.4	5.00	16.4	5.00
Static k=6	40.5	6.00	10.9	6.00	15.1	6.00
Static k=7	36.7	7.00	10.3	7.00	13.7	7.00
Adaptive tau=0.995	65.1	2.09	12.2	2.90	20.2	1.70
Adaptive tau=0.99	64.4	2.31	12.0	3.21	20.0	1.81
Adaptive tau=0.98	63.7	2.52	11.7	3.62	19.8	1.92
Adaptive tau=0.97	62.8	2.73	11.5	3.93	19.6	1.98
Adaptive tau=0.96	62.4	2.85	11.3	4.25	19.5	2.05
Adaptive tau=0.95	61.4	3.06	11.0	4.56	19.4	2.11
Adaptive tau=0.9	58.8	3.57	10.7	4.97	19.0	2.27
Adaptive tau=0.87	56.9	3.88	10.4	5.28	18.7	2.38
Adaptive tau=0.85	56.2	4.00	10.2	5.50	18.5	2.48
Adaptive tau=0.8	54.2	4.31	9.9	5.81	18.1	2.61
Adaptive tau=0.75	52.0	4.62	9.6	6.12	17.8	2.74
Adaptive tau=0.7	49.7	4.93	9.4	6.33	17.3	2.88
Adaptive tau=0.65	46.5	5.35	9.2	6.55	16.8	3.05
Adaptive tau=0.6	43.8	5.66	9.0	6.76	16.1	3.26

Table 8: Complete static and confidence-adaptive decoding sweep for MTP with attraction-only drift (M3). Accuracy is reported in percent; k_{eff} is the measured acceleration factor. All values are synthetic placeholders.

Decoding setting	GSM8K		AIME25		GPQA Main	
	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}
Static k=1	66.6	1.00	12.6	1.00	20.6	1.00
Static k=2	61.7	2.00	12.2	2.00	20.1	2.00
Static k=3	54.5	3.00	11.7	3.00	18.5	3.00
Static k=4	51.7	4.00	11.2	4.00	17.0	4.00
Static k=5	43.9	5.00	10.4	5.00	15.4	5.00
Static k=6	39.3	6.00	9.6	6.00	13.9	6.00
Static k=7	35.3	7.00	8.9	7.00	12.3	7.00
Adaptive tau=0.995	64.9	2.04	11.8	2.84	20.0	1.64
Adaptive tau=0.99	64.3	2.25	11.6	3.15	19.8	1.75
Adaptive tau=0.98	63.5	2.46	11.3	3.56	19.6	1.86
Adaptive tau=0.97	62.7	2.67	11.0	3.87	19.5	1.92
Adaptive tau=0.96	62.2	2.78	10.8	4.18	19.3	1.98
Adaptive tau=0.95	61.3	2.99	10.5	4.49	19.2	2.04
Adaptive tau=0.9	58.7	3.50	10.1	4.90	18.8	2.20
Adaptive tau=0.87	56.9	3.81	9.8	5.21	18.6	2.31
Adaptive tau=0.85	56.2	3.92	9.6	5.42	18.3	2.40
Adaptive tau=0.8	54.2	4.23	9.3	5.73	18.0	2.53
Adaptive tau=0.75	52.1	4.54	9.0	6.04	17.6	2.66
Adaptive tau=0.7	49.8	4.85	8.7	6.25	17.2	2.80
Adaptive tau=0.65	46.5	5.25	8.5	6.45	16.7	2.95
Adaptive tau=0.6	44.0	5.56	8.3	6.66	16.0	3.16

Table 9: Complete static and confidence-adaptive decoding sweep for MTP with block-level full drift (M4). Accuracy is reported in percent; k_{eff} is the measured acceleration factor. All values are synthetic placeholders.

Decoding setting	GSM8K		AIME25		GPQA Main	
	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}
Static k=1	66.9	1.00	12.9	1.00	20.9	1.00
Static k=2	62.4	2.00	12.8	2.00	20.6	2.00
Static k=3	55.4	3.00	12.6	3.00	19.4	3.00
Static k=4	53.0	4.00	12.4	4.00	18.1	4.00
Static k=5	45.4	5.00	11.9	5.00	16.9	5.00
Static k=6	41.2	6.00	11.4	6.00	15.6	6.00
Static k=7	37.4	7.00	11.0	7.00	14.4	7.00
Adaptive tau=0.995	65.1	2.13	12.4	2.93	20.2	1.73
Adaptive tau=0.99	64.5	2.35	12.2	3.25	20.0	1.85
Adaptive tau=0.98	63.7	2.56	11.9	3.66	19.8	1.96
Adaptive tau=0.97	62.8	2.78	11.7	3.98	19.7	2.03
Adaptive tau=0.96	62.3	2.89	11.5	4.29	19.6	2.09
Adaptive tau=0.95	61.4	3.11	11.3	4.61	19.4	2.16
Adaptive tau=0.9	58.7	3.63	11.0	5.03	19.0	2.33
Adaptive tau=0.87	56.9	3.94	10.7	5.34	18.8	2.44
Adaptive tau=0.85	56.1	4.06	10.5	5.56	18.5	2.54
Adaptive tau=0.8	54.1	4.37	10.2	5.87	18.2	2.67
Adaptive tau=0.75	51.9	4.69	10.0	6.19	17.8	2.81
Adaptive tau=0.7	49.5	5.00	9.8	6.40	17.3	2.95
Adaptive tau=0.65	46.2	5.42	9.6	6.62	16.8	3.12
Adaptive tau=0.6	43.6	5.73	9.4	6.83	16.1	3.33

Table 10: Complete static and confidence-adaptive decoding sweep for Soft on-policy KL (M5). Accuracy is reported in percent; k_{eff} is the measured acceleration factor. All values are synthetic placeholders.

Decoding setting	GSM8K		AIME25		GPQA Main	
	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}	Acc. (%)	k_{eff}
Static k=1	66.2	1.00	12.2	1.00	20.2	1.00
Static k=2	61.1	2.00	11.5	2.00	19.3	2.00
Static k=3	53.5	3.00	10.7	3.00	17.5	3.00
Static k=4	50.5	4.00	9.8	4.00	15.6	4.00
Static k=5	42.3	5.00	8.8	5.00	13.8	5.00
Static k=6	37.5	6.00	7.7	6.00	11.9	6.00
Static k=7	33.1	7.00	6.7	7.00	10.1	7.00
Adaptive tau=0.995	64.7	1.94	11.2	2.74	19.6	1.54
Adaptive tau=0.99	64.0	2.14	10.9	3.04	19.5	1.64
Adaptive tau=0.98	63.3	2.34	10.6	3.44	19.3	1.74
Adaptive tau=0.97	62.5	2.55	10.3	3.75	19.2	1.80
Adaptive tau=0.96	62.1	2.65	10.0	4.05	19.1	1.85
Adaptive tau=0.95	61.2	2.85	9.7	4.35	19.0	1.90
Adaptive tau=0.9	58.7	3.35	9.2	4.75	18.6	2.05
Adaptive tau=0.87	56.9	3.66	8.9	5.06	18.4	2.16
Adaptive tau=0.85	56.3	3.76	8.7	5.26	18.1	2.24
Adaptive tau=0.8	54.4	4.06	8.3	5.56	17.8	2.36
Adaptive tau=0.75	52.3	4.36	7.9	5.86	17.5	2.48
Adaptive tau=0.7	50.1	4.67	7.7	6.07	17.1	2.62
Adaptive tau=0.65	47.0	5.07	7.4	6.27	16.6	2.77
Adaptive tau=0.6	44.5	5.37	7.2	6.47	15.9	2.97